

Technical Documentation

Chapter 7: AI Chatbot Implementation

7.1 Architecture

The chatbot operates entirely offline using NLP (Natural Language Processing) and Extractive QA, strictly bypassing LLMs (Large Language Models) to ensure zero internet dependence and rapid execution.

7.2 Processing Workflow

1. Document Upload: Admin uploads a PDF/DOCX.
2. Text Extraction: `PyPDF2`` or `python-docx`` extracts raw text.
3. Chunking: The text is split via Regex by sentence boundaries (``.\! ?``), grouped into blocks of ~150 words. Stored in ``document_chunks``.
4. Querying: Student submits a question.
5. Intent Classification: Basic classification checks for intents like ``EXAM_SCHEDULE`` or ``TIMETABLE`` to route to SQL queries directly.
6. Vectorization (TF-IDF): If intent is general, the system uses Scikit-Learn ``TfidfVectorizer`` to convert all document chunks + the user query into numerical vectors.
7. Cosine Similarity: Measures the cosine angle between the query vector and chunk vectors.
8. Heuristic Filter: Penalizes chunks with high density of '?' to avoid extracting Tables of Contents.
9. Extraction: The highest-scoring chunk is returned to the user neatly formatted with blockquotes.

Chapter 8: API Documentation

Base URL: ``http://localhost:5000/api/v1``

Auth Routes

- ``POST /auth/login``: Accepts JSON ``{username, password}``. Sets HTTP-only session cookie.
- ``GET /auth/me``: Returns current user role/ID.
- ``POST /auth/logout``: Clears session.

Admin Routes (Requires Admin Role)

- ``GET /admin/dashboard``: Returns analytics (total_students, ai_trends, etc).
- ``GET, POST, DELETE /admin/students``: Manage students. POST auto-generates Roll No and creates User.
- ``GET, POST, DELETE /admin/departments``: Manage departments.
- ``GET, POST, DELETE /admin/subjects``: Manage subjects.
- ``GET, POST, DELETE /admin/materials``: Manage materials. POST accepts ``multipart/form-data`` and triggers ``process_and_store_document()``.
- ``GET, POST, DELETE /admin/assignments``: Manage assignments.
- ``GET, POST, DELETE /admin/timetable``: Manage timetables.

Student Routes (Requires Student Role)

- `GET /student/profile`: Get student details.
- `GET /student/timetable`: Get timetable for student's course/semester.
- `GET /student/materials`: Get materials for student's subjects.
- `GET /student/assignments`: Get assignments for student's subjects.